
Insider Threats

A technical presentation of unsupervised anomaly detection for user behavior within the Organization





Problem Statement

- *Business Challenge:* Insider threats (e.g., data exfiltration, system sabotage) pose significant risks due to legitimate access and difficulty in detection.
- *ML Task:* Unsupervised Anomaly Detection. Goal is to identify deviations from 'normal' user behavior without labeled examples of malicious activity.
- *Target:* Flag users exhibiting multi-source anomalies.



Data Overview

Source data: logon events, removable-media device events, and file access logs from CERT synthetic datasets.

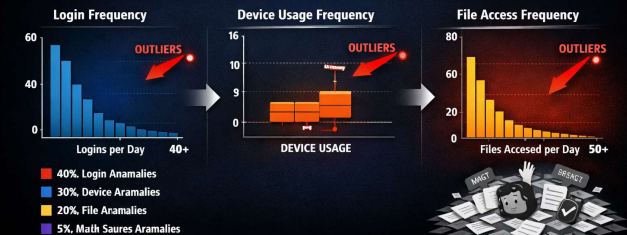
Preprocessing: timestamp alignment, ID standardization, deduplication, and filling missing aggregated feature values with zeros to indicate no activity.

EDA Findings

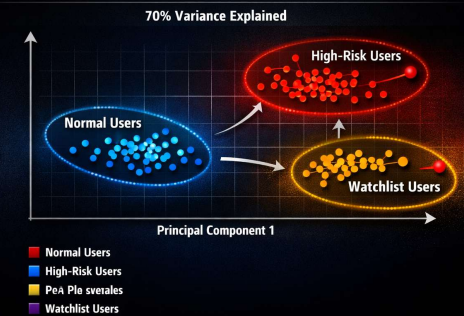
Exploratory analysis revealed skewed feature distributions and clear outliers in *login_frequency*, *file_access_frequency*, and device usage.

PCA captured ~70% variance in two components. Correlations show *login_frequency* strongly relates to *unique_pc_count*, indicating multi-machine access patterns.

FEATURE DISTRIBUTIONS & OUTLIERS



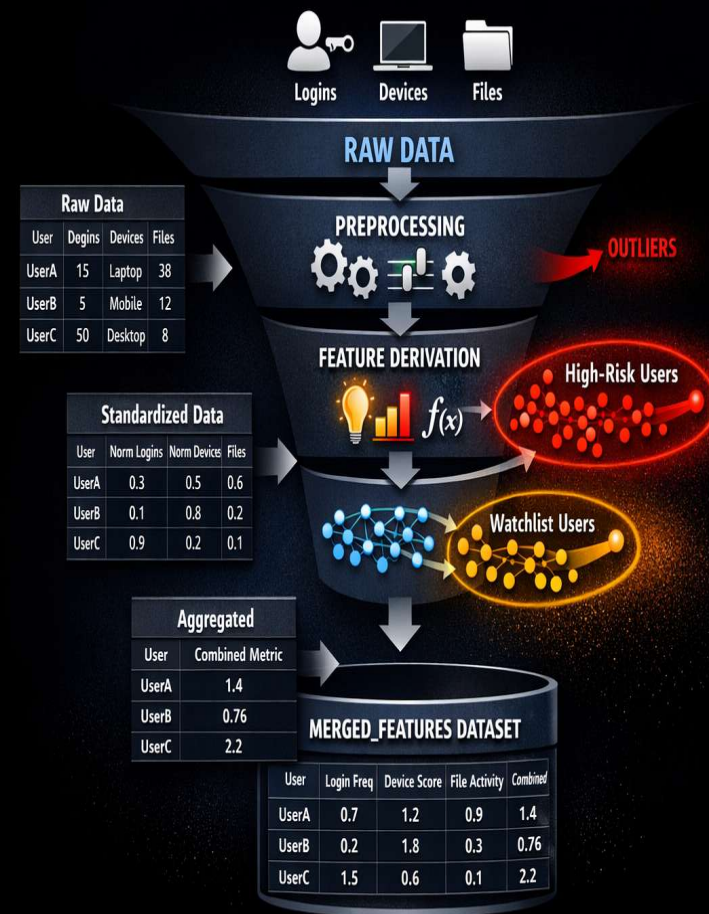
PCA: DIMENSIONALITY REDUCTION



Feature Engineering Approach

Derived temporal features: *date_day*, *hour*, *day_of_week* and behavioral flags like *is_after_hours* and *is_usb_activity*.

Aggregations produced *login_frequency*, *unique_pc_count*, *device_usage_frequency*, and *file_access_frequency*, merged into the *merged_features* dataset with missing values filled as zeros.

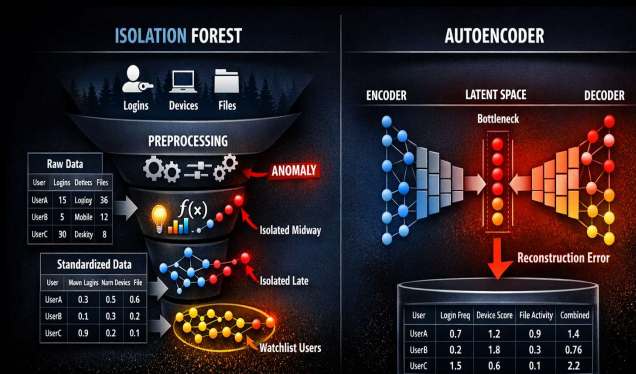


Model Selection & Methodology

Anomaly Detection Approach

- *Unsupervised Learning*: No prior labels for 'malicious' activities exist, so models learn patterns of 'normal' behavior and identify deviations.
- *Multi-Source Integration*: Models are trained on a unified dataset (merged_features) combining features from logon, device, and file access data.
- *Employed unsupervised methods*: *Isolation Forest* ($n_estimators=300$, contamination tuned from 0.5 to 0.07) and an *Autoencoder* (32-16-8 bottleneck).

Models trained on the unified features to detect deviations from learned normal behavior and flag potential insider threats.



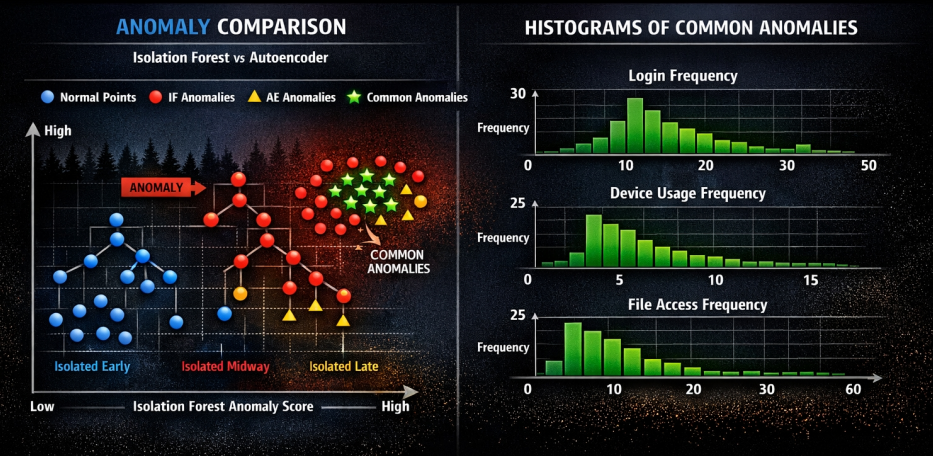
Results & Model Comparison

Used SHAP for Isolation Forest to identify drivers like `login_frequency` and `unique_pc_count`.

Autoencoder anomalies interpreted via reconstruction error (threshold = $\text{mean MSE} + 2\sigma$) to pinpoint contributing features for targeted investigations.

Overall Agreement: High (~95%) between IF (initial) and AE, indicating general consistency in anomaly vs. normal classifications.

Common Anomalies: Number of commonly detected anomalies significantly increased when IF contamination is set from 0.5 to 0.7



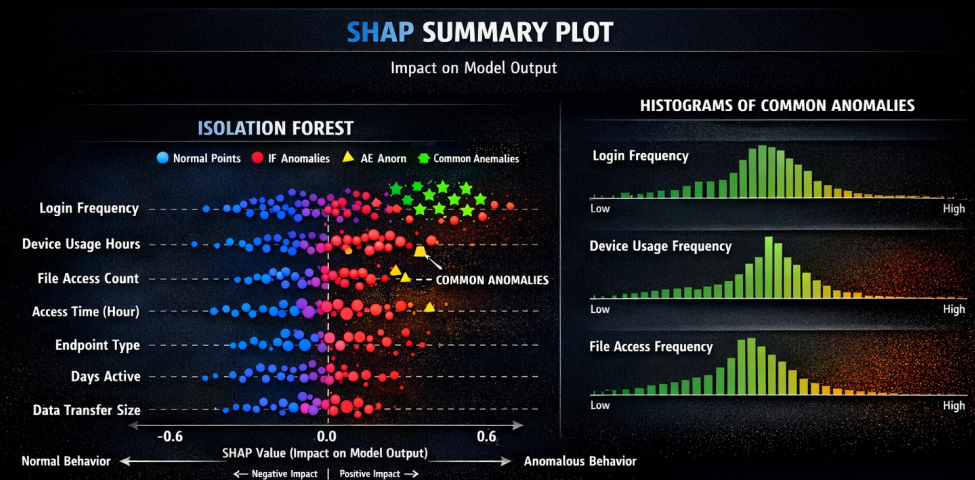
Feature Importance & Explainability

Understanding Anomaly Drivers

- Goal: To move beyond simply flagging anomalies to understanding why specific user activities are deemed anomalous.
- Methodology: Utilizing explainable AI techniques to interpret model predictions.

Key Findings (from SHAP Summary Plot):

- login_frequency and unique_pc_count were consistently the most influential features. High values for login_frequency and unique_pc_count often significantly increased the likelihood of an instance being flagged as an anomaly, suggesting unusual login patterns or access to multiple machines. Low values in these features were generally associated with normal behavior.
- file_access_frequency and device_usage_frequency also showed significant importance. High values in these features, especially removable_media_ratio, tended to push predictions towards anomalous, indicating suspicious data handling or device connections.



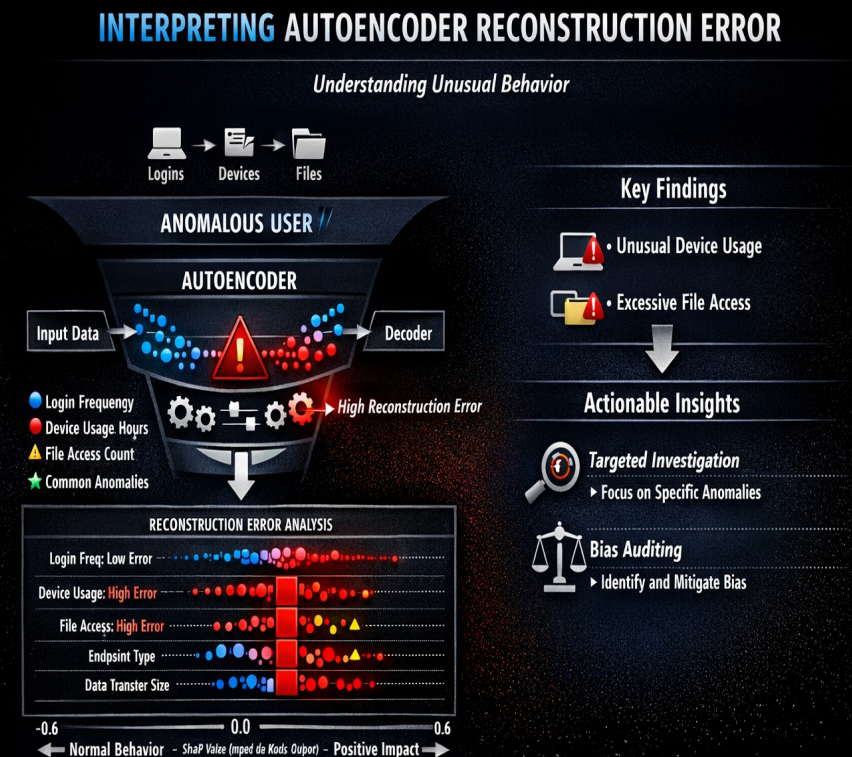
Feature Importance & Explainability (cont.)

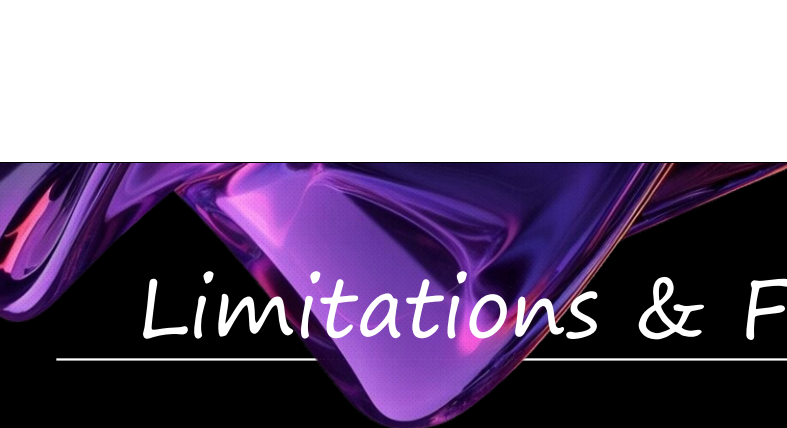
Interpreting Autoencoder Reconstruction Error

- **Mechanism:** A high reconstruction error from the Autoencoder indicates that the model struggled to reconstruct the input, suggesting the input pattern is 'unseen' or 'unusual'.
- **Explainability:** By examining the features within an instance that contributed most to the high reconstruction error, we can infer which specific aspects of the user's behavior deviated significantly from the learned 'normal' patterns.
- For example, an unexpectedly high device_usage_frequency or file_access_frequency for a user's typical behavior would result in a high reconstruction error for those specific features within an anomalous instance. This points security teams directly to the unusual activity.

Actionable Insights

- **Targeted Investigation:** Explainability allows security teams to focus investigations on the specific behavioral deviations highlighted by the models.
- **Bias Auditing:** Understanding feature influence can also help in identifying and mitigating potential biases in anomaly detection, ensuring fairness across user groups.





Limitations & Future Work

Model Limitations

- *Class Imbalance:* Anomalies are inherently rare, leading to models potentially overfitting to the majority 'normal' class and missing subtle threats.
- *Data Leakage Risk:* Careful feature engineering is required to prevent features from different datasets inadvertently leaking information about each other, which could lead to overoptimistic performance.
- *Generalization Challenges:* Models trained on synthetic CERT data may not fully generalize to the unique behavioral patterns and complexities of real-world organizational environments. On the other hand, the model could pick up biases from real world data, if they exist.
- *Proxy Bias:* Features like 'user' or 'PC' could inadvertently act as proxies for sensitive attributes, leading to biased anomaly detection rates across different demographic groups or teams.

Areas for Future Work

- *Real-time Detection:* Implement a streaming architecture to enable near real-time anomaly detection, allowing for quicker response to emerging threats.
- *Integration of More Data Sources:* Incorporate additional data, such as HR records, network traffic logs, or communication data, to enrich behavioral profiles and enhance detection accuracy.
- *Advanced Explainability:* Further explore advanced XAI techniques to provide more granular, human-readable explanations for why a specific event or user behavior was flagged as anomalous.
- *Feedback Loop Integration:* Develop a system to incorporate feedback from security analysts to continuously refine model performance and adapt to evolving threat landscapes.
- *Semi-Supervised Learning:* Investigate semi-supervised approaches if limited labeled anomaly data becomes available, to potentially improve model precision and recall.